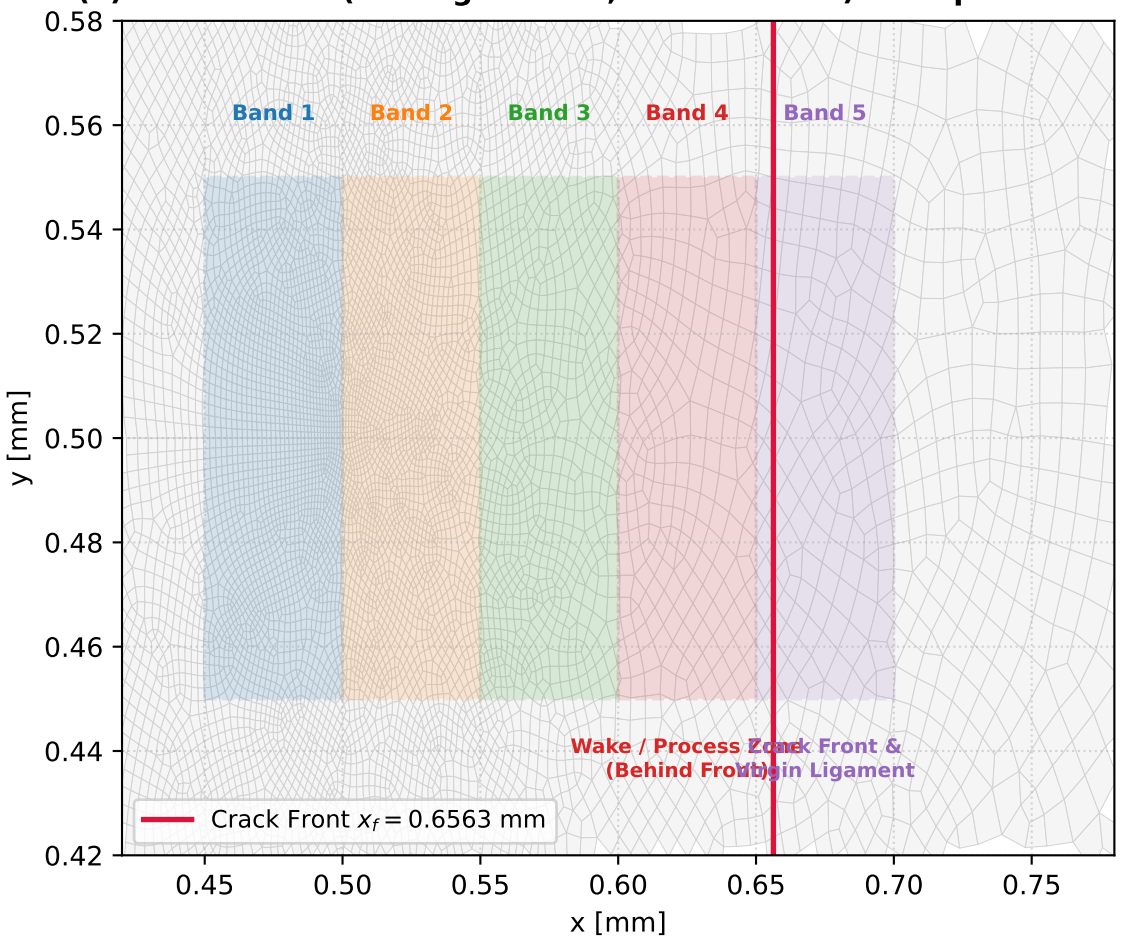
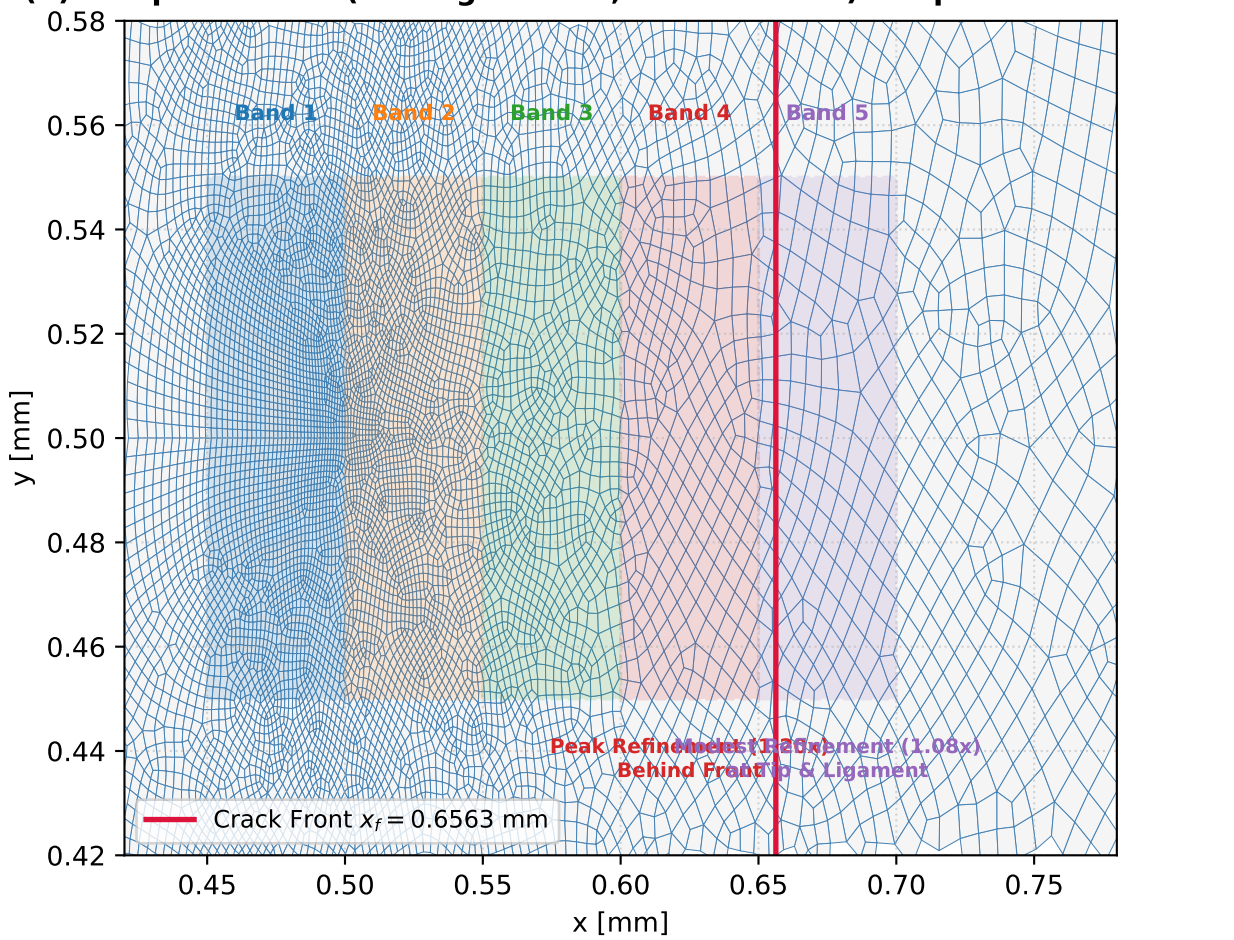
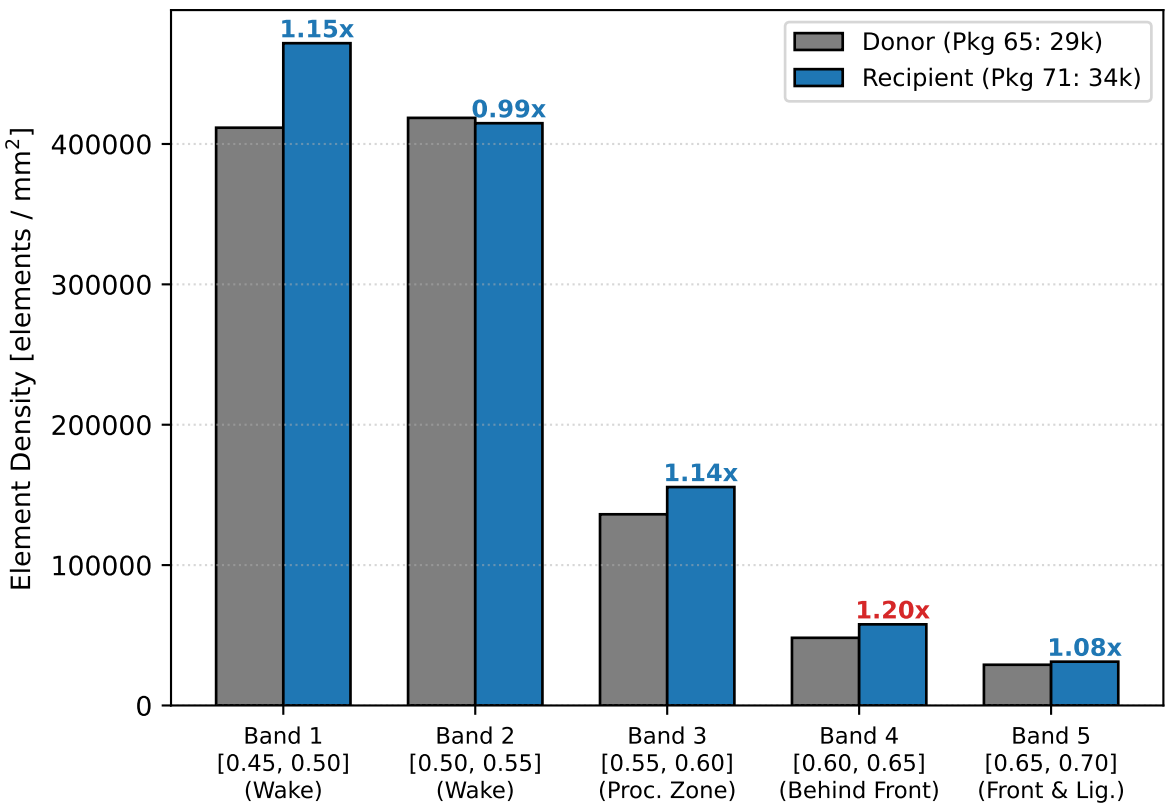






(d) Quantitative Mesh & Density Metrics Across Spatial Bands

Band & Physical Ro	x-Range [mm	Donor Elem	Recip Elem	Refine Ratio	Donor $\bar{h}$ [μm	Recip $\bar{h}$ [μm
Band 1 (Wake)	[0.45, 0.50]	2058	2359	1.15x	1.5	1.4
Band 2 (Wake)	[0.50, 0.55]	2093	2074	0.99x	1.5	1.5
Band 3 (Proc. Zone)	[0.55, 0.60]	681	778	1.14x	2.6	2.5
Band 4 (Behind Front (x <	[0.60, 0.65]	241	289	1.20x	4.5	4.1
Band 5 (Front & Ligamer	[0.65, 0.70]	145	156	1.08x	5.7	5.6